Stefano Carlo **Lambertenghi**

Doctoral Candidate @ TUM / Scientific Researcher @ fortiss GmbH

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Summary

Doctoral candidate at the Technical University of Munich specializing in autonomous driving and robotics. Experienced in deploying and validating both modular and end to end autonomous driving systems on real vehicles, with hands on work in trajectory planning, ROS based ADS stacks, camera and LiDAR perception, neural rendering, and simulation to real transfer.

Work Experience

fortiss GmbH. *München, Germany*
Scientific Researcher *Jun. 2023 - Now*

- Responsible for the development, deployment, operation, and maintenance of the Fortuna autonomous research vehicle, a Volkswagen Passat adapted by IAV and certified for autonomous driving on public roads. Restored the platform to full operational status and currently lead its use for autonomous driving research and testing.
- Designed and conducted nearly 100 hours of autonomous driving experiments on public roads. Deployed and adapted Autoware and Apollo, including modifications to trajectory planning for narrow two way urban roads and experience with lattice based planners. Also deployed end to end driving models from NVIDIA DAVE-2 to recent CARLA Leaderboard agents and developed methods for real world validation and sim to real evaluation.

IDSIA, Dalle Molle Institute for Artificial Intelligence *Lugano, Switzerland*
Intern *Sep. 2022 - Jun. 2023*

- Developed ultra low power deep learning based monocular relative localization for nano quadrotors, enabling real time onboard inference on resource constrained embedded hardware.

eVs embedded Vision Systems s.r.l. *Verona, Italy*
Intern *May. 2019 - Aug. 2019*

- Developed FPGA based vehicle classification algorithms using Support Vector Machines for intelligent guidance systems.

Significant Publications

Real-World Perturbation Testing of Autonomous Driving Systems *fortiss GmbH - TUM*
ASE 2026 *2026*

- Performed the first large scale real world evaluation of 72 camera and LiDAR perturbations on modular and end to end autonomous driving systems deployed on a Volkswagen Passat, demonstrating that commonly used offline robustness metrics poorly predict real world system performance.

Cam2Sim: Neural Scenario Reconstruction for Closed-Loop Autonomous Driving Simulation *fortiss GmbH - TUM*
ASE 2026 *2026*

- Developed *Cam2Sim*, a framework that reconstructs ROS driving recordings into executable CARLA scenarios using Gaussian Splatting, substantially improving visual realism and enabling closed loop evaluation from real world data.

Misbehavior Forecasting for Focused Autonomous Driving Systems Testing *fortiss GmbH - TUM*
ICSE 2026 *2026*

- Introduced *Foresee*, a forecasting framework that predicts near future driving failures and guides search based fuzzing toward safety critical scenarios, significantly increasing testing efficiency.

A Multi Modality Evaluation of the Reality Gap in Autonomous Driving Systems *fortiss GmbH - TUM*
ASE 2025 *2025*

- Compared Software in the Loop, Vehicle in the Loop, Mixed Reality, and real world testing, providing the first systematic study of their complementary strengths and limitations for autonomous driving validation.

Benchmarking Image Perturbations for Testing Automated Driving Assistance Systems *fortiss GmbH - TUM*
ICST 2025 – **IEEE TCSE Distinguished Paper Award** *2025*

- Established the first comprehensive benchmark of 38 image perturbation categories for testing camera based ADAS, showing that perturbation driven augmentation significantly improves model robustness under adverse visual conditions.

Ultra Low Power Deep Learning based Monocular Relative Localization Onboard Nano Quadrotors *IDSIA*
ICRA 2023 *2023*

- Developed an ultra low power deep learning localization pipeline for nano drones, achieving 48 Hz onboard inference at only 95 mW.

Education

TUM (Technical University of Munich) *Munich, Germany*
Doctoral Candidate *Jun 2023 - Now*

- Topic: Measuring and mitigating the sim to real gap in testing of cyber physical systems.

USI (Università della Svizzera Italiana) *Lugano, Switzerland*
Master of Science in Artificial Intelligence *Oct. 2020 - Sep. 2022*

- Thesis: Drone-to-drone on-board relative pose estimation and control of Nano-class autonomous UAVs.
- GPA: 9.21, **Summa cum laude**

UNIVR (Università di Verona) *Verona, Italy*
Bachelor's degree in IT science and technology *Oct. 2016 - Jul. 2020*

- Thesis: Interactive Holographic displays for advertisement.
- GPA: 97.

Skills

Programming languages Python, C/C++/C#, Java, SWIFT, MATLAB
Framework and tools PyTorch, TensorFlow, OpenCV, Pandas, NumPy, ROS, Docker, Unity, Blender, Autoware, Apollo
Languages Italian (Native), English (C1, TOEFL iBT English certificate, 115/120)